LTE Air Interface

Course objectives:

The LTE Air Interface course offers a complete understanding of the radio interface in Long Term Evolution (LTE) technology. A general characteristic of LTE is presented with stress on its advantages and disadvantages. The course focuses on the physical channels structure and their processing in both FDD and TDD mode of LTE. The course also presents procedures of power control, link adaptation and HARQ related to downlink and uplink scheduling. The concept of MIMO in LTE, idle mode behavior and handover process are also discussed. The course is based on the 3GPP Technical Specifications and hence is equipment vendor independent.

Duration Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to managerial staff and network engineers involved in the radio network planning and optimization process, as well as to anybody who requires detailed knowledge on the LTE radio interface structure and function.

Course Contents:

- OFDMA principles
  - FDD and TDD)
  - OFDM / OFDMA and SC-OFDM
  - OFDM transmitter
  - OFDM receiver
  - Modulation

- EPS architecture
  - EPS architectural principles (EPC and E-UTRAN)
  - Strata (NAS and AS)

- LTE signaling overview
  - User plane
  - Control plane
  - Protocols (RRC, PDCP, RLC, MAC)
  - Radio interface structure

- LTE radio interface introduction
  - Channel structure (logical channels, transport channels and physical channels)
  - Time domain structure for FDD and TDD
  - Frequency domain structure
  - Scheduling Block
  - Virtual Resource Block (localized and distributed type)
  - System spectral efficiency

- LTE downlink physical channels
  - Cell search
  - P-SS
  - S-SS
  - RS
  - PBCH (MIB and SIB)
  - PCFICH
  - PDCCH (usage, mapping, format, processing, blind decoding)
  - PDSCH (CRC attachment, code block segmentation, channel coding, rate matching, code block concatenation, scrambling, modulation mapper, layer mapper, precoding, resource element mapping)
  - PHICH
  - PMCH
  - Downlink physical channels modulation

- LTE uplink physical channels
  - PUSCH
  - PUCCH
  - PRACH
  - SRS

- Physical layer procedures
  - Timing advance (uplink-downlink frame timing, timing advance range, random access, maintenance of uplink time alignment)
  - Random Access
  - Resource Allocation (Type 0, 1 and 2)
  - MIMO (spatial multiplexing, transmit diversity, transmission modes, MIMO antennas)
  - UE reporting (CQI definition, aperiodic CQI/PMI/RI reporting using PUSCH, periodic CQI/PMI/RI reporting using PUCCH)
  - Modulation order and transport block size determination
  - UL power control (PUSCH power control, PUSCH power control example).

- LTE mobility
  - Idle mode mobility (PLMN selection, cell selection, cell reselection)
  - Connected mode mobility (X2 handover, event triggered reporting)

Pre-requisites:

Training participants should bring a deep understanding of the EPC architecture, nodes and functions.
Alternatively, prior to this training, participants should attend the course LTESysO.

Course Code: LTEAirIF

Price per student:

For Option 1, Classroom training : USD 1,800.00 per student

For Options 2, 3, Virtual Classroom training: USD 1,500.00 per student